EXPERIMENT NO. 6

Food Delivery Application – Positive Testing

Question

Write a case study to identify and analyze the requirements of Food Delivery Applications, and prepare positive test scenarios as well as test cases for the application.

Aim

To analyze the functional requirements of a food delivery application and prepare positive test scenarios and test cases using valid inputs.

Requirements Analysis

- Users should be able to register and log in with valid details.
- Users should be able to search restaurants, view menus, and add available food items to the cart.
- The application should calculate item totals, taxes, delivery charges, and discounts correctly.
- Users should be able to enter a valid delivery address, select a payment method, and place an order.
- The system should display order confirmation and allow users to track order status.

Algorithm / Procedure

1. Study the application requirements and identify the main user functions.
2. Prepare valid input data such as an active account, available food items, a valid address, and an accepted payment method.
3. Define positive scenarios for login, search, cart, checkout, payment, and tracking.
4. Execute each test case and compare the actual output with the expected output.
5. Record the actual result and mark each case as Pass or Fail after execution.

Test Scenarios

Scenario ID	Scenario Description	Type
FD-P01	Login with valid credentials	Positive
FD-P02	Search for an available restaurant or food item	Positive
FD-P03	Add available food items to cart and update quantity	Positive
FD-P04	Apply a valid coupon and calculate total	Positive
FD-P05	Place an order using a valid address and successful payment	Positive
FD-P06	View order confirmation and track an order	Positive

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Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
FD-TC01	Open login page; enter registered email and correct password; select Login.	User logs in and reaches the home page.	To be recorded	Not Executed
FD-TC02	Enter a valid food name, such as “Veg Pizza,” in search.	Matching available items or restaurants are displayed.	To be recorded	Not Executed
FD-TC03	Select an available item; add it to cart; increase quantity from 1 to 2.	Cart displays the correct item and quantity, and updates the subtotal.	To be recorded	Not Executed
FD-TC04	Enter a valid coupon code for an eligible order and apply it.	Discount is applied according to coupon rules and the total is recalculated.	To be recorded	Not Executed
FD-TC05	Choose a valid saved address, select an available payment method, and confirm order.	Order is created, payment is confirmed, and an order ID is displayed.	To be recorded	Not Executed
FD-TC06	Open the placed order from Order History.	Correct order details and current status are displayed.	To be recorded	Not Executed

Result

The positive test scenarios and test cases for the Food Delivery Application were prepared. Actual results and final Pass/Fail status can be entered after executing the tests.

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EXPERIMENT NO. 7

Food Delivery Application – Negative Testing

Question

Write a case study to identify and analyze the requirements of Food Delivery Applications, and prepare negative test scenarios as well as test cases for the application.

Aim

To verify that the food delivery application handles invalid inputs, unavailable services, and unsuccessful transactions safely.

Requirements Analysis

- Mandatory registration and login fields must be validated.
- Unavailable menu items must not be ordered.
- Invalid coupons and unsupported delivery addresses must be rejected with clear messages.
- Payment failures must not create a confirmed paid order.
- Repeated clicks or retries must not create duplicate orders or duplicate charges.

Algorithm / Procedure

6. Identify fields and operations where invalid input or failure can occur.
7. Prepare invalid credentials, empty fields, expired coupons, unavailable items, and failed-payment conditions.
8. Execute each negative test and observe the validation message or recovery behavior.
9. Check that invalid actions do not change cart, order, or payment data incorrectly.
10. Record actual behavior and report any unexpected result as a defect.

Test Scenarios

Scenario ID	Scenario Description	Type
FD-N01	Submit login form with an incorrect password	Negative
FD-N02	Attempt checkout with a missing delivery address	Negative
FD-N03	Apply an expired or invalid coupon	Negative
FD-N04	Attempt to order an item that is out of stock	Negative
FD-N05	Simulate payment failure or timeout	Negative
FD-N06	Click Place Order repeatedly during checkout	Negative

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
FD-TC01	Enter a registered email with an incorrect password and select Login.	Login is denied and a suitable error message is displayed; the user remains logged out.	To be recorded	Not Executed
FD-TC02	Add an item to cart; proceed to checkout without selecting or	Checkout is blocked and the address field is identified as	To be recorded	Not Executed

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	entering an address.	required.		
FD-TC03	Enter an expired or invalid coupon code and apply it.	Coupon is rejected, an explanatory message is displayed, and the total remains correct.	To be recorded	Not Executed
FD-TC04	Select an item marked unavailable or make it unavailable before checkout.	The application prevents ordering the item and informs the user of its availability.	To be recorded	Not Executed
FD-TC05	Use a test payment method that returns failure or timeout.	Payment is not marked successful; the user receives a failure message and no paid order is created.	To be recorded	Not Executed
FD-TC06	Click the order confirmation control multiple times quickly.	Only one order is created and the customer is not charged more than once.	To be recorded	Not Executed

Result

The negative test scenarios and test cases were prepared to check the application's validation and error-handling behavior. Execution is required to confirm the actual results.

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EXPERIMENT NO. 8

Banking Application – Positive Testing

Question

Write a case study to identify and analyze the requirements of Banking Applications, and prepare positive test scenarios as well as test cases for the application.

Aim

To identify banking application requirements and prepare positive test cases for valid authentication, balance enquiry, fund transfer, and transaction history.

Requirements Analysis

- Customers must be able to log in using valid credentials and required authentication.
- Account balances and transaction details must be displayed accurately.
- Fund transfers must validate beneficiary details and available balance.
- Successful transactions must update balances correctly and generate a reference number.
- Sensitive banking operations must be recorded securely.

Algorithm / Procedure

11. Review the banking functions and their expected outcomes.
12. Prepare valid test data using approved test accounts and beneficiaries.
13. Test login, balance enquiry, beneficiary selection, fund transfer, and transaction history.
14. Verify that debit and credit entries are consistent after a successful transfer.
15. Record the actual results without using real customer funds or credentials.

Test Scenarios

Scenario ID	Scenario Description	Type
BK-P01	Login using valid credentials and authentication	Positive
BK-P02	Check the balance of an active account	Positive
BK-P03	Transfer an allowed amount to a registered beneficiary	Positive
BK-P04	View recent transaction history	Positive
BK-P05	Download or view a transaction receipt	Positive

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
BK-TC01	Enter a valid test username, password, and required OTP; select Login.	Customer is authenticated and the account dashboard is displayed.	To be recorded	Not Executed
BK-TC02	Open Balance Enquiry for an active account.	The current available balance is displayed accurately.	To be recorded	Not Executed

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BK-TC03	Select a registered beneficiary; enter an amount within the permitted limit; confirm with valid authentication.	Transfer succeeds once, source balance decreases, beneficiary balance increases, and a reference number is generated.	To be recorded	Not Executed
BK-TC04	Open Transaction History and choose a valid date range.	Transactions within the selected range are displayed with correct date, amount, and status.	To be recorded	Not Executed
BK-TC05	Open a completed test transaction and select View Receipt.	A receipt with the correct reference number, amount, date, and transaction status is displayed or downloaded.	To be recorded	Not Executed

Result

The positive test scenarios and test cases for the Banking Application were prepared to validate normal banking operations. The cases should be executed only in a controlled test environment.

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EXPERIMENT NO. 9

Banking Application – Negative Testing

Question

Write a case study to identify and analyze the requirements of Banking Applications, and prepare negative test scenarios as well as test cases for the application.

Aim

To test how the banking application rejects invalid credentials and transactions while protecting account data and funds.

Requirements Analysis

- The system must reject incorrect login details and invalid OTPs.
- Transfers must be blocked when the balance is insufficient or the amount exceeds transaction limits.
- Invalid beneficiary details and malformed amounts must be rejected.
- Repeated requests must not result in duplicate transfers.
- Session timeout and access-control rules must protect customer information.

Algorithm / Procedure

16. Identify risky banking operations and validation rules.
17. Prepare invalid credentials, incorrect OTP, insufficient-balance accounts, invalid amounts, and unregistered beneficiaries.
18. Run each test in a test environment and observe the system response.
19. Verify that rejected transactions do not debit or credit accounts.
20. Document any incorrect balance change, missing validation, or data exposure as a defect.

Test Scenarios

Scenario ID	Scenario Description	Type
BK-N01	Log in with an incorrect password or OTP	Negative
BK-N02	Transfer an amount greater than the available balance	Negative
BK-N03	Transfer to an unregistered or invalid beneficiary	Negative
BK-N04	Enter zero, a negative amount, or invalid characters in amount field	Negative
BK-N05	Submit the same transfer request more than once	Negative
BK-N06	Attempt to access account information after session expiry	Negative

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
BK-TC01	Enter an incorrect password or	Access is denied; an	To be recorded	Not Executed

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	incorrect OTP and attempt login.	appropriate message is displayed and security controls are applied.		
BK-TC02	Use a test account with a known balance and enter a transfer amount greater than that balance.	Transfer is rejected and neither account balance changes.	To be recorded	Not Executed
BK-TC03	Enter invalid beneficiary details and confirm a test transfer.	The system rejects the beneficiary and does not initiate the transfer.	To be recorded	Not Executed
BK-TC04	Enter 0, a negative number, or alphabetic characters in the transfer amount field.	Input validation blocks the request and explains the acceptable amount format.	To be recorded	Not Executed
BK-TC05	Resubmit the same transfer reference/request in the test environment.	The application processes the transfer at most once and prevents duplicate debit or credit.	To be recorded	Not Executed
BK-TC06	Allow the banking session to expire, then try to open account details.	The user is redirected to authentication and protected account details are not exposed.	To be recorded	Not Executed

Result

The negative test scenarios and test cases were prepared to verify transaction validation, access control, and protection against duplicate transfers. Actual execution is needed to confirm the system behavior.

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EXPERIMENT NO. 10

Comprehensive Test Plan – Library Management System

Question

Develop a comprehensive Test Plan document for a chosen application, such as a Library Management System, outlining various testing strategies, methodologies, and criteria to ensure thorough evaluation and validation of the system's functionality.

Aim

To develop a structured test plan for validating the major functions, quality attributes, and operational readiness of a Library Management System.

Requirements Analysis

- Scope: user login, book search, book issue/return, renewals, reservations, fine calculation, member management, and reports.
- Out of scope: third-party payment provider internals and unrelated external systems.
- Test levels: unit testing, integration testing, system testing, and user acceptance testing.
- Test types: functional, usability, compatibility, security, performance, and regression testing.
- Resources: test environment, browser/device combinations, sample members, books, and transaction records.

Algorithm / Procedure

21. Analyze requirements and identify features included in and excluded from testing.
22. Prepare a requirement traceability matrix and prioritize high-risk functions.
23. Design test cases using equivalence partitioning, boundary value analysis, decision tables, and state transition testing.
24. Prepare test data and configure a test environment that resembles production without using real private data.
25. Execute smoke tests, functional tests, integration tests, security checks, and regression tests.
26. Log defects, retest fixes, perform regression testing, and prepare a final test summary.

Testing Strategy and Methodology

- Black-box testing will verify outputs against requirements without relying on internal code.
- Risk-based testing will prioritize login, issue/return, fine calculation, and member data protection.
- Regression testing will be performed after fixes or changes to critical functions.
- Exploratory testing will be used to identify usability problems and unexpected workflows.

Test Design Techniques

- Equivalence partitioning: group valid and invalid member IDs or search inputs.
- Boundary value analysis: test loan limits, due dates, fine thresholds, and maximum reservation counts.
- Decision table testing: check combinations of member status, book availability, and borrowing limit.
- State transition testing: verify book states such as Available, Issued, Reserved, and Returned.

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Execution Strategy

- Run smoke tests after each deployment before detailed testing.
- Execute test cases in priority order and capture evidence for failures.
- Report defects with ID, steps to reproduce, expected result, actual result, severity, and screenshots where applicable.
- Retest corrected defects and run regression tests on related modules.

Entry and Exit Criteria

- Entry: requirements are reviewed, test build is available, environment is configured, and test data is prepared.
- Exit: all critical test cases have been executed, no unresolved critical defects remain, high-priority defects are addressed or formally accepted, and the test summary is approved.
- Completion criteria such as acceptable pass percentage and response time must be agreed by the project team before execution.

Test Scenarios

Scenario ID	Scenario Description	Type
LIB-01	Member login and role permissions	Functional / Security
LIB-02	Search books by title, author, or ISBN	Functional
LIB-03	Issue and return a book; update availability	Functional / Integration
LIB-04	Calculate overdue fine and process renewal	Boundary / Functional
LIB-05	Performance during concurrent searches	Performance
LIB-06	Verify usability and supported browsers	Usability / Compatibility

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
LIB-TC01	Log in with a valid member account and open the member dashboard.	Dashboard opens and only permitted member functions are available.	To be recorded	Not Executed
LIB-TC02	Search using an existing title, author, and ISBN.	Correct matching books and availability status are displayed.	To be recorded	Not Executed
LIB-TC03	Issue an available book to an eligible member, then return it.	Issue/return records are saved and availability is updated correctly.	To be recorded	Not Executed
LIB-TC04	Return a book after its due date and verify the fine calculation against configured rules.	Fine matches the approved fine rule; renewal and payment status are updated as specified.	To be recorded	Not Executed
LIB-TC05	Run the agreed load test with concurrent searches and measure response time and errors.	The application meets the performance targets defined by the project team.	To be recorded	Not Executed
LIB-TC06	Repeat critical functions on supported browsers and screen sizes.	Pages remain usable and key functions work consistently on supported configurations.	To be recorded	Not Executed

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Result

The test plan for the Library Management System was prepared, covering scope, methods, execution, and completion criteria. Final acceptance depends on actual test results and approved project thresholds.

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EXPERIMENT NO. 11

Railway Reservation System – Positive Testing

Question

Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases for the application.

Aim

To validate normal reservation workflows using valid passenger details, available trains, and successful payment.

Requirements Analysis

- Users should be able to search trains by source, destination, and journey date.
- Available classes and seat counts must be displayed correctly.
- Valid passenger details and eligibility information must be accepted.
- Successful payment should generate a confirmed booking and PNR/reference number.
- Users should be able to view booking details and cancel according to applicable rules.

Algorithm / Procedure

27. Review search, passenger, seat availability, payment, and booking confirmation requirements.
28. Prepare valid source/destination, future journey date, passenger details, and test payment data.
29. Search trains and select a train with available seats.
30. Complete passenger details and payment in the test environment.
31. Verify booking confirmation, PNR, fare, and booking history.

Test Scenarios

Scenario ID	Scenario Description	Type
RR-P01	Search a train for a valid route and future date	Positive
RR-P02	Select an available train and class	Positive
RR-P03	Enter valid passenger details and continue	Positive
RR-P04	Complete payment for an available booking	Positive
RR-P05	View confirmed booking and PNR	Positive

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
RR-TC01	Enter valid source, destination, and future journey date; select Search.	Matching trains and schedules are displayed.	To be recorded	Not Executed
RR-TC02	Select a train/class showing available seats.	Selected train, class, fare, and availability are shown correctly.	To be recorded	Not Executed
RR-TC03	Enter valid passenger name, age, and required	Passenger details are accepted and the booking proceeds to	To be recorded	Not Executed

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	identity/contact details.	payment.		
RR-TC04	Complete payment using a valid test payment method.	Payment succeeds and the reservation is confirmed once.	To be recorded	Not Executed
RR-TC05	Open booking history after confirmation.	Correct journey details, passenger information, fare, booking status, and PNR are displayed.	To be recorded	Not Executed

Result

The positive test scenarios and test cases for the Railway Reservation System were prepared. They can be executed in a test environment to verify successful reservation workflows.

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EXPERIMENT NO. 12

Railway Reservation System – Negative Testing

Question

Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases for the application.

Aim

To check that the railway reservation system rejects invalid searches and passenger data and handles unavailable seats or failed payments correctly.

Requirements Analysis

- Source and destination must be valid and should satisfy route rules.
- Journey dates must follow the application's booking policy.
- Required passenger fields must be validated.
- Bookings must not exceed availability or create duplicate reservations.
- Failed or timed-out payments must not leave an incorrect confirmed booking.

Algorithm / Procedure

32. Identify invalid and boundary conditions in train search and booking.
33. Prepare invalid route/date combinations, incomplete passenger details, unavailable seats, and failed-payment conditions.
34. Execute each negative test and capture the displayed error or recovery behavior.
35. Verify that no incorrect booking, seat deduction, or duplicate payment occurs.
36. Document unexpected behavior as a defect for correction and retesting.

Test Scenarios

Scenario ID	Scenario Description	Type
RR-N01	Search with identical source and destination or an invalid station	Negative
RR-N02	Enter a date outside the permitted booking window or an invalid past date	Negative
RR-N03	Submit passenger details with required fields missing or invalid	Negative
RR-N04	Attempt booking when no seats are available	Negative
RR-N05	Simulate payment failure or timeout	Negative
RR-N06	Submit the same booking request more than once	Negative

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
RR-TC01	Enter an invalid station or use	Search is rejected or a clear	To be recorded	Not Executed

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	the same source and destination, then search.	validation message is shown; no invalid route is offered.		
RR-TC02	Choose a past date or a date outside the configured booking window.	The system blocks the search/booking and explains the valid date range.	To be recorded	Not Executed
RR-TC03	Leave a mandatory passenger field blank or enter an invalid value.	The field is highlighted and booking cannot proceed until corrected.	To be recorded	Not Executed
RR-TC04	Select a train/class with zero availability and attempt confirmation.	The system prevents confirmation or offers permitted alternatives without overselling seats.	To be recorded	Not Executed
RR-TC05	Use a test payment method that fails or times out.	Booking remains unconfirmed or enters a clearly defined pending state; payment status is handled safely.	To be recorded	Not Executed
RR-TC06	Send repeated booking submissions for the same request.	Only one valid reservation is created and seat availability is not deducted more than once.	To be recorded	Not Executed

Result

The negative test scenarios and test cases were prepared to validate error handling and prevent invalid or duplicate railway reservations.

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EXPERIMENT NO. 13

Online Examination System – Positive Testing

Question

Write a case study to identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases for the application.

Aim

To verify that eligible students can log in, access an assigned examination, answer questions, submit responses, and view results when permitted.

Requirements Analysis

- Registered students must be able to authenticate and view assigned exams.
- Exams must open only within the configured date and time window.
- Questions, options, marks, and timer must be displayed correctly.
- Answers should be saved as students progress through the exam.
- Submission must be recorded once and results must follow the exam's release policy.

Algorithm / Procedure

37. Review student authentication, exam scheduling, answering, autosave, submission, and result requirements.
38. Prepare an eligible student account and an active test exam.
39. Log in, open the exam during the permitted period, and answer sample questions.
40. Verify saved answers, submission confirmation, and result visibility based on settings.
41. Record actual behavior and compare it with the expected outcome.

Test Scenarios

Scenario ID	Scenario Description	Type
OE-P01	Log in using a registered student account	Positive
OE-P02	Open an assigned exam during the active exam window	Positive
OE-P03	Answer questions and navigate between them	Positive
OE-P04	Submit the completed examination	Positive
OE-P05	View result after results are released	Positive

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
OE-TC01	Enter valid student credentials and select Login.	Student dashboard is displayed with the assigned examination list.	To be recorded	Not Executed
OE-TC02	Select an assigned exam during its active time window.	Exam instructions, questions, and timer are displayed	To be recorded	Not Executed

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		correctly.		
OE-TC03	Select answers, move between questions, and return to a previous question.	Answers are retained and displayed correctly during navigation.	To be recorded	Not Executed
OE-TC04	Answer the required questions and select Submit; confirm when prompted.	Responses are saved and a submission confirmation is displayed.	To be recorded	Not Executed
OE-TC05	Open the results page after the instructor releases results.	The correct result and permitted feedback are displayed for the student.	To be recorded	Not Executed

Result

The positive test scenarios and test cases for the Online Examination System were prepared to validate the expected examination workflow.

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EXPERIMENT NO. 14

Online Examination System – Negative Testing

Question

Write a case study to identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases for the application.

Aim

To test the system's response to invalid login details, unauthorized exam access, missing responses, timing violations, and submission errors.

Requirements Analysis

- Invalid credentials must not grant access.
- Students must not access exams that are not assigned to them or are outside the permitted window.
- Required answers and input formats must be validated according to exam rules.
- Expired timers and repeated submissions must be handled consistently.
- One student must not access another student's exam or result.

Algorithm / Procedure

42. Identify access, timing, input, and submission rules.
43. Prepare invalid accounts, unassigned exams, closed exams, incomplete answers, and expired-session conditions.
44. Execute the negative scenarios and verify the error messages and access restrictions.
45. Check that answers and submissions are not lost or duplicated unexpectedly.
46. Record failures and retest after correction.

Test Scenarios

Scenario ID	Scenario Description	Type
OE-N01	Log in with invalid credentials	Negative
OE-N02	Attempt to open an unassigned or closed examination	Negative
OE-N03	Attempt to submit with required answers missing where answers are mandatory	Negative
OE-N04	Continue using an expired exam session or after time expires	Negative
OE-N05	Submit the same exam repeatedly	Negative
OE-N06	Attempt to view another student's exam or result	Negative

Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
OE-TC01	Enter an incorrect password or unknown student ID and select	Access is denied and a suitable error message is displayed.	To be recorded	Not Executed

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	Login.			
OE-TC02	Open an exam that is not assigned to the student or whose window has closed.	The system denies access and displays the appropriate availability message.	To be recorded	Not Executed
OE-TC03	Leave a mandatory question unanswered and select Submit.	The system follows the configured rule: prompts the student to answer or clearly warns before submission.	To be recorded	Not Executed
OE-TC04	Allow the timer/session to expire and attempt to continue answering.	The exam is handled according to the configured timeout policy and unauthorized extra time is not granted.	To be recorded	Not Executed
OE-TC05	Click Submit repeatedly or resend a submission request.	Only one final submission is recorded; duplicate attempts do not alter the final answer set.	To be recorded	Not Executed
OE-TC06	Change a student identifier or URL to request another student's result.	Access is denied and private exam/result data is not disclosed.	To be recorded	Not Executed

Result

The negative test scenarios and test cases were prepared to check examination security, timing rules, validation, and reliable submission.

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EXPERIMENT NO. 15

Comprehensive Test Plan – Inventory Management System

Question

Develop a comprehensive Test Plan document for an Inventory Management System, covering test scope, test design techniques, test execution strategy, and test completion criteria.

Aim

To create a complete test plan for verifying inventory records, stock movements, supplier information, alerts, and reporting functions.

Requirements Analysis

- Scope includes product creation/editing, stock-in, stock-out, stock adjustment, supplier management, reorder alerts, search, and reports.
- Out of scope includes external supplier systems not controlled by the application and physical warehouse procedures.
- Test levels include unit, integration, system, regression, and user acceptance testing.
- Quality goals include data accuracy, access control, usability, compatibility, and performance.
- Test data includes products, categories, suppliers, stock quantities, reorder levels, and sample purchase/sales transactions.

Algorithm / Procedure

47. Analyze requirements and define the scope, assumptions, dependencies, and risks.
48. Identify roles such as administrator, inventory staff, and report viewer, and map their permissions.
49. Design tests using equivalence partitioning, boundary value analysis, decision tables, and state transitions.
50. Prepare a clean test database and realistic product and transaction data.
51. Run smoke tests, functional tests, integration tests, role/security checks, performance tests, and regression tests.
52. Log defects, retest fixes, confirm stock calculations, and produce a test completion report.

Test Design Techniques

- Equivalence partitioning: group valid and invalid product IDs, SKU values, and quantities.
- Boundary value analysis: test stock quantity at zero, one unit, reorder level, and maximum permitted values.
- Decision table testing: combine stock level, reorder threshold, and product status to validate alert behavior.
- State transition testing: verify product stock states such as In Stock, Low Stock, and Out of Stock.

Test Execution Strategy

- Prioritize stock adjustment, stock-out prevention, and transaction consistency because errors affect business records.
- Execute smoke tests after deployment and functional tests before integration and regression suites.
- Test integration between product records, stock movements, supplier receipts, alerts, and reports.
- Maintain evidence such as screenshots, transaction IDs, logs, and defect records.

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Entry and Exit Criteria

- Entry: requirements are approved, test build is available, test environment and data are ready, and major blockers are resolved.
- Exit: all critical test cases are executed, no unresolved critical defects remain, stock calculations reconcile with transaction history, and remaining risks are documented and approved.
- Agree on measurable performance targets and required pass percentage with the project team before test execution.

Defect Reporting

- Record defect ID, summary, environment, preconditions, steps to reproduce, expected result, actual result, severity, priority, and supporting evidence.
- After a fix, rerun the failed test and perform regression testing on related stock and reporting functions.

Test Scenarios

Scenario ID	Scenario Description	Type
INV-01	Create product and search by SKU/name	Functional
INV-02	Record stock received from a supplier	Functional / Integration
INV-03	Record stock issue or sale and recalculate stock	Functional
INV-04	Test zero, negative, and boundary stock quantities	Boundary / Negative
INV-05	Verify reorder alert at configured threshold	Decision Table
INV-06	Verify roles, reports, and concurrent updates	Security / Performance

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Test Cases

Test Case ID	Test Case / Input and Steps	Expected Result	Actual Result	Status
INV-TC01	Create a product with a unique SKU, valid name, category, and reorder level; search for it.	Product is saved and appears in search with correct details.	To be recorded	Not Executed
INV-TC02	Record receipt of a known quantity for an existing product.	Available stock increases by exactly the received quantity and the movement is recorded.	To be recorded	Not Executed
INV-TC03	Issue a valid quantity that does not exceed available stock.	Stock decreases correctly and the transaction history records the issue.	To be recorded	Not Executed
INV-TC04	Try entering zero, a negative quantity, or a quantity greater than available stock for a stock issue.	Invalid quantity is rejected and stock remains unchanged.	To be recorded	Not Executed
INV-TC05	Reduce stock to the configured reorder level and refresh the inventory dashboard.	The reorder alert appears according to the configured threshold rule.	To be recorded	Not Executed
INV-TC06	Log in using different roles and attempt permitted and restricted actions; run concurrent stock updates.	Each role can access only authorized functions, and stock values remain consistent after updates.	To be recorded	Not Executed

Result

The test plan for the Inventory Management System was prepared with scope, test design techniques, execution strategy, and completion criteria. Final completion should be declared only after test evidence is reviewed and exit criteria are satisfied.

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